

LaunchPad AI – Smart Business Idea Evaluation System

Abstract:

LaunchPad AI is an AI-powered web application designed to help entrepreneurs evaluate and validate their startup ideas before launching them. The system uses Google Gemini AI to analyze startup details provided by users and generates comprehensive reports, including Startup Analysis, SWOT Analysis, Market Analysis, and Business Recommendations. Developed using Python and Flask, the application offers a simple and interactive interface that enables users to receive AI-generated business insights instantly. By integrating Generative AI with a web application, LaunchPad AI supports entrepreneurs in making informed business decisions and improving their startup ideas.

Problem Statement:

Many entrepreneurs and students have innovative startup ideas but lack the knowledge and resources to evaluate their feasibility. Traditional business consulting is often expensive and time-consuming, making it difficult for individuals to obtain expert guidance. There is a need for an intelligent AI-powered system that can quickly analyze startup ideas and provide meaningful business insights, helping users identify strengths, weaknesses, opportunities, risks, and possible improvements before investing time and resources..

Features:

- AI-powered startup idea evaluation.
- User-friendly web interface for entering startup details.
- Generates Startup Analysis.
- Generates SWOT Analysis.
- Provides Market Analysis.
- Suggests Business Recommendations.
- Stores previous analyses in the History page.
- Responsive frontend developed using HTML, CSS, and JavaScript.
- Fast AI analysis through Flask backend.
- Cloud deployment using Render.
- Modular and maintainable project structure.

Tech Stack:

Type of Tech Stack	Technologies Used
Frontend	HTML,CSS, JavaScript

Backend	Python, Flask
Database	SQLite, SQLAlchemy
AI Model	Google Gemini 2.5 Flash
Development Tools	Visual Studio Code, Git, GitHub
Deployment	Render, Gunicorn
Environment	Python-dotenv

Installation Steps:

1. Install Python 3.10 or above.
2. Clone or download the project repository.
3. Open the project folder in Visual Studio Code.
4. Install all required dependencies using:
 `pip install -r requirements.txt`
5. Create a `.env` file and add:
 `GEMINI_API_KEY=your_api_key`
 `SECRET_KEY=your_secret_key`
6. Run the Flask application:
 `python app.py`

Project Directory Structure

The project directory was organized as:

LaunchPadAI/

|— app.py

|— config.py

|— requirements.txt

|— Procfile

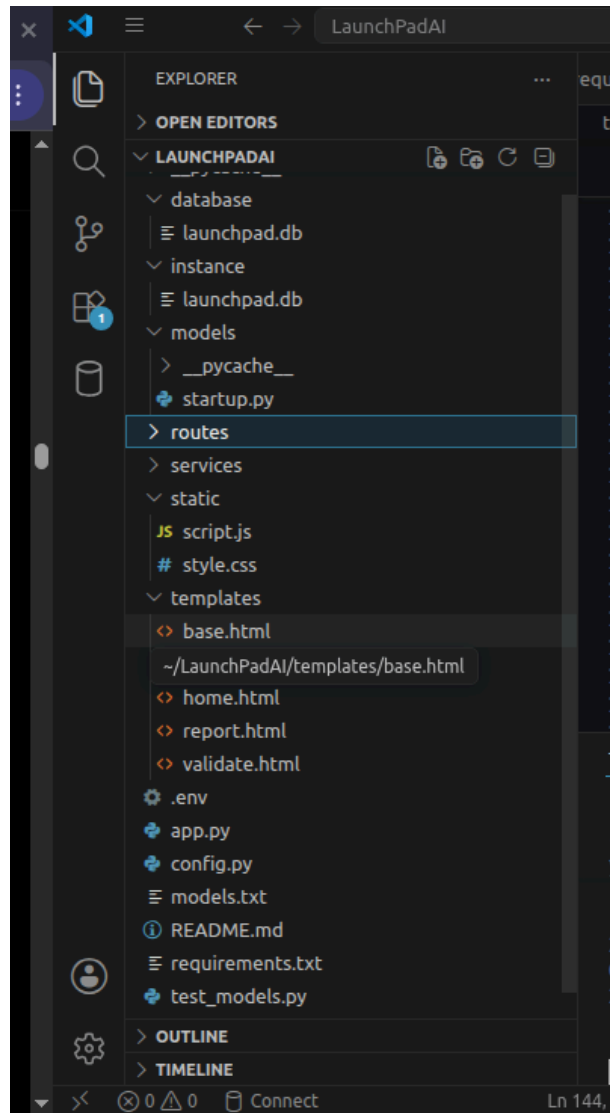
|— README.md

|— .env.example

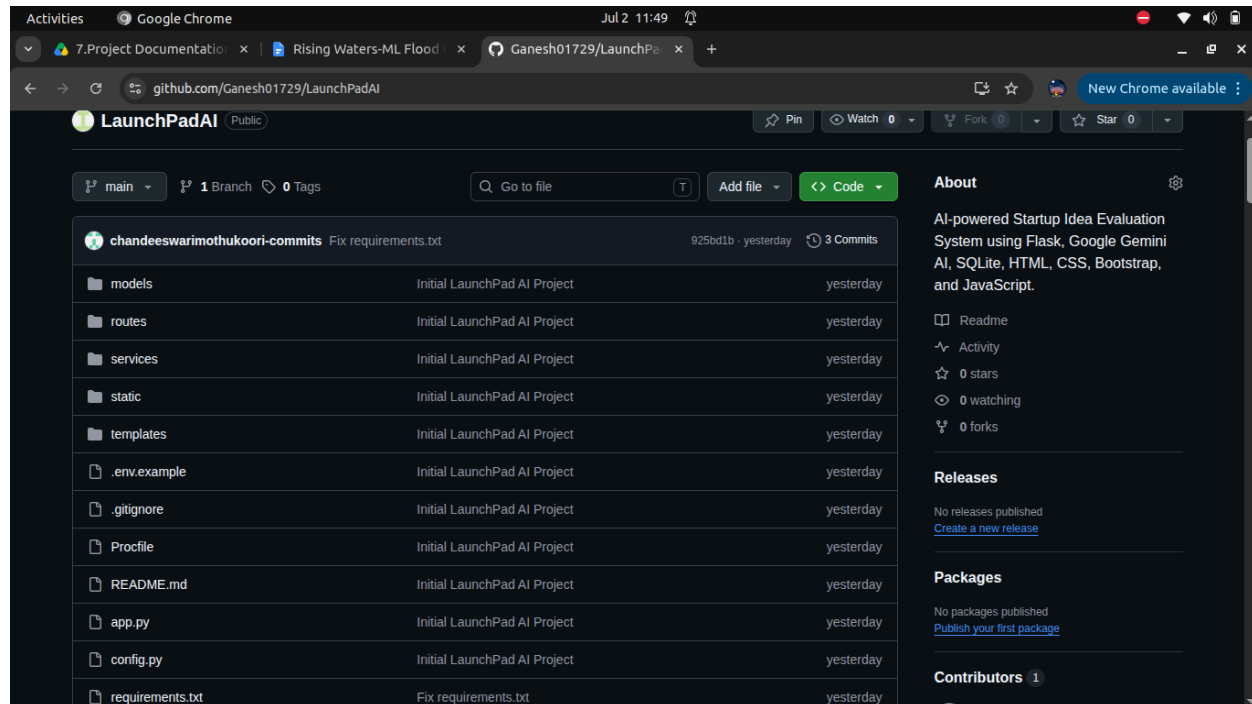
|

|— models/

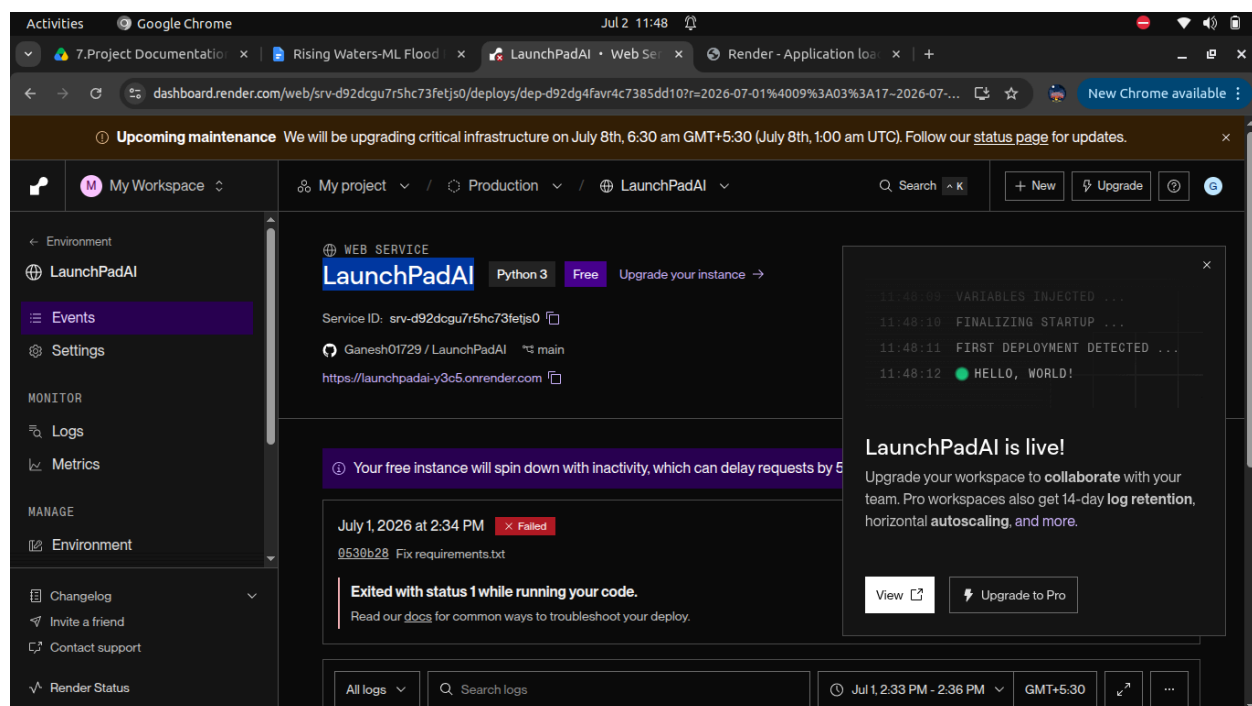
```
| └─ startup.py
|
|─ routes/
|   └─ home.py
|   └─ history.py
|
|─ services/
|   └─ gemini_service.py
|   └─ prompt.py
|
|─ templates/
|   └─ base.html
|   └─ home.html
|   └─ validate.html
|   └─ report.html
|   └─ history.html
|
|─ static/
|   └─ style.css
|   └─ script.js
|
└─ launchpad.db
```



Git connection to Render



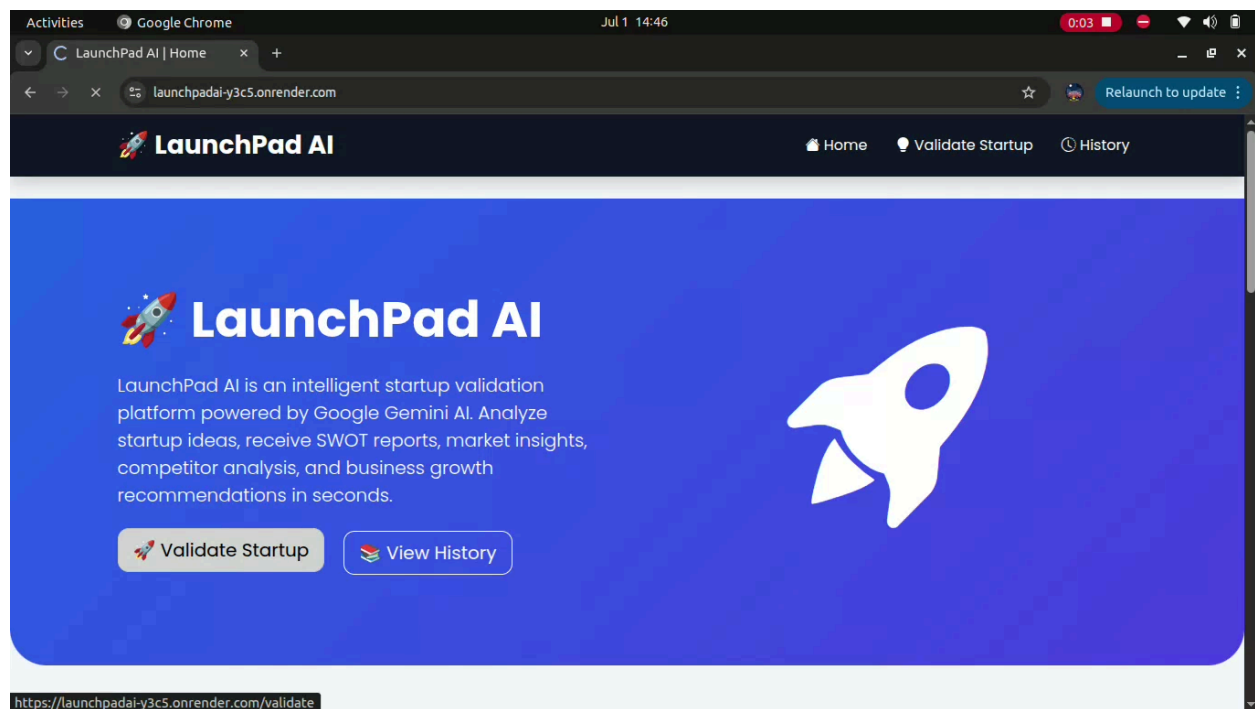
Render Dashboard



Usage Instructions:

- Launch the application in a web browser.
- Open the Startup Evaluation page.
- Enter the Startup Name, Startup Idea, Target Audience, Business Model, and Product Features.
- Click the **Analyze Startup** button.
- The application sends the details to Google Gemini AI.
- View the generated Startup Analysis, SWOT Analysis, Market Analysis, and Business Recommendations.
- Open the History page to review previous analyses.

Live Website after Deployment



Validate Startup | Launch x +

launchpadai-y3c5.onrender.com/validate

Relaunch to update

 **LaunchPad AI**

Home Validate Startup History

 **LaunchPad AI**

Enter your startup details and let **Google Gemini AI** generate a professional startup analysis.

You'll Receive

✓ Startup Validation Score

✓ SWOT Analysis

✓ Market Analysis

✓ Competitor Analysis

✓ Revenue Strategy

✓ Customer Acquisition

Startup Submission

Startup Name

Eco cart

Startup Idea

A

Minimum 50 characters | Maximum 2000 characters

1 / 2000 characters

Target Audience

Validate Startup | Launch x +

launchpadai-y3c5.onrender.com/validate

Relaunch to update

✓ Revenue Strategy

✓ Customer Acquisition

✓ Growth Roadmap

Minimum 50 characters | Maximum 2000 characters

53 / 2000 characters

Target Audience

Urban Customers

Business Model

Subscription

Key Product Features

AI recommendat

Analyze Startup

Startup Analysis Report

launchpadai-y3c5.onrender.com/report/2

Relaunch to update


LaunchPad AI

HomeValidate StartupHistory



Startup Analysis Report

AI Generated Business Intelligence using Google Gemini AI


Startup Details

Startup Name Eco cart	Target Audience Urban Customers
Business Model Subscription	Generated On 01 July 2026 09:18 AM

Viability


Innovation


Market


Execution


Startup Analysis Report

launchpadai-y3c5.onrender.com/report/2

Relaunch to update

- Subscription model provides predictable recurring revenue.
- "Fast delivery" addresses a key customer pain point in urban environments.
- Potential for strong customer loyalty due to personalized service and local connection.

Weaknesses

- Complex logistics due to coordinating pick-ups from multiple, potentially disparate local stores.
- Onboarding and maintaining partnerships with numerous independent local stores can be resource-intensive.
- Inventory management across various local stores without centralized control is challenging.
- High initial capital investment required for AI development, delivery infrastructure, and market penetration.
- Potential for higher operating costs compared to centralized warehouse models.

Opportunities

- Growing consumer preference for supporting local businesses and sustainable sourcing.
- Increasing demand for convenient, on-demand delivery services in urban areas.
- Expansion into specialized local products (e.g., artisanal goods, organic produce, baked goods).
- Leveraging AI to optimize delivery routes, personalize deals, and predict demand for local stores.
- Partnerships with local community events, food festivals, and apartment complexes.

Team Member Details:

S.No	Team Member Name	Role	Responsibilities
1	Ganesh Naik (Team Leader)	Full Stack Developer & AI Integration	Independently designed and developed the complete LaunchPad AI – Smart Business Idea Evaluation System, including frontend development, Flask backend, Google Gemini AI integration, SQLite database management, testing, GitHub version control, documentation, and deployment on Render.

Future Scope:

The **LaunchPad AI – Smart Business Idea Evaluation System** can be further enhanced by incorporating additional features and advanced AI technologies to improve analysis quality, usability, and scalability. Some potential future improvements include:

- Integrate real-time market and industry data for more accurate startup evaluation.
- Add competitor analysis using online business and market information.
- Generate complete business plans and investor-ready pitch decks automatically.
- Include financial forecasting and revenue estimation features.
- Develop an AI-powered chatbot to provide startup guidance and answer user queries.
- Enable PDF export and report download for generated startup analyses.
- Store user accounts and startup history using cloud databases such as MySQL or PostgreSQL.
- Support multiple languages to make the platform accessible to users worldwide.
- Deploy the application using Docker and Kubernetes for better scalability and reliability.
- Develop Android and iOS mobile applications for easy access on smartphones.
- Integrate with cloud platforms such as IBM Cloud, AWS, or Azure for improved performance and availability.
- Continuously improve AI prompts and models to provide more accurate and personalized business recommendations.

Conclusion:

The **LaunchPad AI – Smart Business Idea Evaluation System** is an AI-powered web application developed using **Flask** that helps entrepreneurs evaluate and validate their startup ideas using **Google Gemini AI**. The system integrates a responsive frontend with an intelligent AI service to provide users with comprehensive **Startup Analysis, SWOT Analysis, Market Analysis, and Business Recommendations**. The project includes frontend development, backend implementation, AI integration, database management, prompt engineering, testing, and cloud deployment using **Render**.

The application enables users to enter startup details such as the startup name, business idea, target audience, business model, and product features to receive detailed AI-generated business insights. The Flask backend processes the user input, communicates with Google Gemini AI, and displays structured reports that help users understand the strengths, weaknesses, opportunities, risks, and improvement suggestions for their startup ideas.

The project demonstrates how **Generative AI** and modern web technologies can be combined to build an intelligent decision-support system for startup evaluation. The deployed application provides easy online access and can be enhanced further with additional business intelligence features.

Future improvements may include integrating real-time market data, competitor analysis, financial forecasting, business plan generation, PDF report export, multilingual support, mobile application development, and deployment on cloud platforms such as IBM Cloud, AWS, or Azure. With continuous improvements in AI models and business analytics, LaunchPad AI can evolve into a comprehensive AI-powered startup consulting platform.

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